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① Intelligent Agents in a Smart Home Automation system :

Introduction :

An intelligent agent is a system that preceive its environment through sensors and acts upon it using actuators to achieve specific objectives. in a smart home automation system sensors collect information such as motion temparature, lightning conditions, and time of day. the agent uses this information to control lights, air conditioning, and security devices while learning user preferences.

1. Types of Intelligent Agents :-

a) simple Reflex agent

A simple Reflex agent selects actions based only on the curr percent using condition-action rules. it does not consider past experiences or maintain an internal model of the environme

Example : it a motion sensor detects a person in a dark ro the agent switches on the lights. if no motion is dete for a specified period, it switches them off.

b) model-Based agent :

A model based agent maintains internal state representing relevant aspects of the enviro it uses current sensor inputs and previous information to upd this state and make decisions.

Example:- The agent remembers that a person is inside the house even when a motion sensor temporarily stops detecting movement. It can also combine temperature readings with door and window sensor information.

c) Goal-Based Agent:-

A goal based agent chooses actions that help achieve a predefined goal. It considers possible actions and their consequences before selecting an appropriate action.

Example:- if the goal is to maintain a comfortable room temperature of 24°c the agent decides whether to switch on the air conditioner, increase ventilation or adjust the thermostat.

Comparision of intelligent Agents:-

Agent type	Decision making Method	Smart home application
Simple reflex	current percept and Rules	motion-based light
Model-based	current input and internal state	occupancy tracking
Goal based	Actions that achieve goals	Temperature control

Most Effective Agent:- Hybrid approach:-

A hybrid agent combining model-based goal based and u
based reasoning is most effective for a smart home au
ion system.

→ The model-based component maintains information about room occupancy, user habits and environmental conditions.

→ The goal based component works toward objectives such as maintaining comfortable temperature and securing house

Conclusion:

A simple reflex agent is suitable for basic automation, while model-based and goal-based agents provide better awareness and planning. A Utility-based hybrid approach is the most effective because it balances comfort energy efficiency personalization and security while responding to changing environmental conditions.

Uniform search strategies in a unknown Maze:

Introduction:

Uniform Search, also called blind sea Explores possible states without using additional k dge about how close a state is to the goal in an unk Maze, a robot can represent each location as a st and each movement - up, down left or right - as an a

1) Uniform cost search (UCS):

Uniform cost search expands the node with the low cummulative path cost, $g(n)$ it uses a priority Queu select the least-cost unexplored node.

Working procedure :

1. place the starting position in a priority queue with cost zero.
2. Remove the node with the lowest cummulative cost.
3. check whether the node represents the exit. if so, return the path.
4. otherwise, explore : valid neighboring cells and calculate their cummulative costs.
5. Add New or cheaper paths to the priority Queue.
6. Repeat until the exit is found or no reachable nodes Remain.

Ex : if one route to the exit-s costs 12 units and another costs 8 units, ucs selects the route costing 8 units, provided all step costs are non-negative and the search completes

Advantages :

→ Handles mazes in which different movements have different costs.

Disadvantages :

→ may explore many states before reaching the exit.

2) Iterative Deepening search (IDS) :

Iterative Deepening search repeatedly performs depth limited depth-first search, increasing the depth limit until the goal is found.

Working procedure :

→ set the depth limit to zero.

→ stop when the exit is reached.

ages:

→ uses much less memory than UCS.

Disadvantages:

→ may take longer when the solution is deep.

comparision of UCS and IDS:

Feature	Uniform cost search	Iterative Deepening Search
Search method	Lowest cumulative cost first	Increasing depth
Data structure	priority Queue	Depth-first search
Optimality	minimum-cost path	shallowest path equal step cost

4) Relationship to intelligent Agents:

Different intelligent agents affects how the robot perceiv the maze and makes decisions.

a) simplex Reflex agent

b) model-Based Agent

c) Goal-Based agent

d) Utility-based agent.

Conclusion :- UCS is suitable for finding minimum-cost paths while IDS is useful when memory is limited and movement costs are equal. combing an appropriate search strategy with model-based and goal based agent improves the robots ability to explore the Maze.